(c) The total number of passengers in the bus are $55 + 11 = 66$. Since each passenger pays $Shs.1,000$, the total amount of money collected is $66 \times 1,000 = Shs.\ 66,000$.

NMC 2015 PRIMARY LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. Norah holds the regional games record for the 200m at 20.2 seconds. If she can improve the record by 2%, what would be her new time?

Qn2. The sum of two numbers is S. Suppose 3 is added to each number and then each of the resulting numbers is doubled. Find the sum of the final two numbers in terms of S.

Qn3. The clock is showing 11am now. When the hour hand has turned through 165^0, what time will it be?

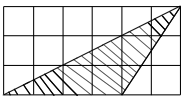
Qn4. Maria is making a special drink by mixing 350ml of water with 150ml of syrup. How much more syrup would she need to add in order to achieve a ratio of syrup to water of 4 : 7?

Qn5. Annet, Ben, Charles, David, Emma and Fred sit around a circular table in that order. Annet starts by saying 2015, Ben says 2016, Charles says 2017, David says 2018, and so on round the table. Who will eventually say 5012?

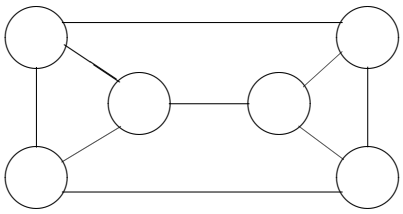
Qn6. At exactly 10am, triplets Claire, Lydia and Alex all sneeze. Claire then sneezes every 20 seconds, Lydia every 30 seconds and Alex every 50 seconds. When will the triplets next sneeze at the same time?

Qn7. In a family, Daddy is 40 years, Mummy is 38 years, their son is 4 years and daughter is 6 years. In how many years' time will their ages total 116?

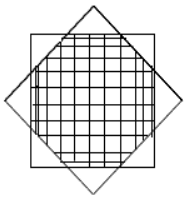
Qn8. What fraction of the rectangle below is shaded?



Qn9. Place the digits $1, 2, 3, 4, 5, 6$ in the circles so that the difference between any two numbers connected by a line is always greater than 1.



Qn10. Two squares each of side 8cm overlap to form the octagon below (not to scale). The area of each triangle is 2cm^2. What is the area of the octagon (shaded portion)?



SECTION B

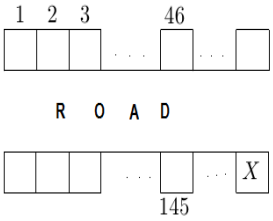
Qn11. A water tank that measures $3 \times 4 \times 5$ (in metres) can be filled by a water pipe in 20 minutes.

(a) What volume of water is in the tank in one minute?

(b) How long will the water pipe take to fill a tank measuring $6 \times 12 \times 10$ (in metres)?

Qn12. A guard and a thief spot one another 100m apart. The thief runs away at a speed of 8km/hr and, at the same time, the guard starts chasing him at a speed of 12km/hr. They both maintain their speed. How long will it take the guard to catch the thief?

Qn13. A multi-billion housing company has an equal number of flats on each side of the road arranged as below:



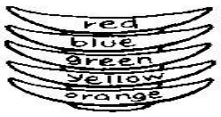
The flats are numbered $1, 2, 3, \ldots$ along one side and continued in this clockwise direction on the other side of the road so that the highest numbered flat is opposite flat number 1. Flat 46 is directly opposite flat 145.

(a) What is the number of the flat marked X?

(b) If instead, the flats are numbered in such a way that odd numbers are all on one side $(1, 3, 5, \ldots)$ and even numbers on the opposite side $(2, 4, 6, \ldots)$ so that flat number 1 is directly opposite flat number 2. Is flat number 46 still directly opposite flat 145? Why?

Qn14. If you write all the whole numbers from 1 to 100 in words, how many more times will you write the letter V than the letter L?

Qn15. There are five coloured plates in a pile arranged as Red, Blue, Green, Yellow, Orange as shown above. Jane takes the top two plates and puts them under the bottom plate (with the red still above the blue). Then again takes the top two plates and puts them at the bottom. If she does this operation a total of 52 times, what colour plate will be on top?



NMC 2015 PRIMARY-LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1. Improvement of running time by $2\% \Longrightarrow$ new time is $(100-2)\% = 98\%$ of the old record. Thus the new time $= 98\% \times 20.2 = 0.98 \times 20.2 = 19.796$sec.

Qn2. Let the numbers be a and b such that $s = a + b$; after adding 3 to both a and b; and doubling the result; the numbers are $2(a+3)$ and $2(b+3)$ and then their sum is $2(a+3) + 2(b+3) = 2a + 6 + 2b + 6 = 2(a+b+6)$. But $a + b = s$; thus the sum of the final numbers in terms of s is $2(s+6) = 2s + 12$.

Qn3. 12 hours on a clock are equivalent to $360^o \Longrightarrow 1$ hour $\equiv \frac{360^o}{12} = 30^o$.

$30^o \equiv 1$ hour $\Longrightarrow 1^o \equiv \frac{1}{30}$ hours. Thus $165^o \equiv \frac{1}{30} \times 165 = 5.5$ hours. Hence from 11am after a 165° turn by the hour hand, the time will be $11 + 5.5 = 16.5$ hours which is $16:30$ hours or $4:30$pm.

Qn4. The ratio of 150ml of syrup to 350 ml of water is 150:350 = 3:7. Total volume = 150+350 = 500ml and from the ratio 3:7; total number of portions = 3+7 = 10. Hence each portion is equivalent to $\frac{500}{10} = 50$ml. Thus 4 parts in the ratio 4:7 will be equivalent to $4 \times 50 = 200$ml. Thus Maria will need an additional 200-150=50ml of syrup to attain a ratio of syrup to water of 4:7.

Qn5. After Annet has said 2015; there are $5012 - 2015 = 2997$ numbers remaining. There are six people thus they will mention $\frac{2997}{6} = 499$ numbers, remainder 3. So the last count is by the 3rd person after Annet and this is David.

Qn6. Sneezes will next coincide at the LCM of $20, 30$ and 50; which is 300. They will sneeze at the same time after 300sec = 5min; thus the time will be $10:05$am.

Qn7. Current total age is $40 + 38 + 4 + 6 = 88$ years. To make a total of 116 it will require an additional $116 - 88 = 28$ years.

Each year the total of the four family members increases by 4. Thus the additional 28 years will be attained in $\frac{28}{4} = 7$ years' time.

Qn8. Area of the rectangle is $3 \times 6 = 18$ squares.

The unshaded upper triangle is half the rectangle; thus its area $= \frac{18}{2} = 9$ squares.

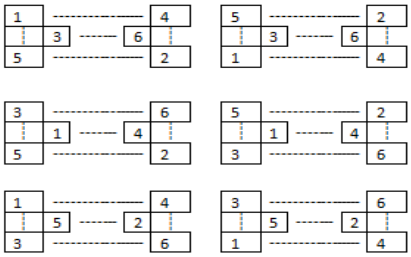
Alternatively its area $= \frac{1}{2}bh = \frac{1}{2} \times 3 \times 6 = 9$ squares.

The area of the unshaded small triangle on the right $= \frac{1}{2} \times 2 \times 3 = 3$ squares.

Thus the area of the shaded triangle $= 18 - 9 - 3 = 6$ squares.

Therefore the fraction of the rectangle that is shaded $= \frac{3}{18} = \frac{1}{3}$.

Qn9. There are various ways of doing this but crucial to note that 1, 3, 5 will always be connected and also 2, 4, 6 to meet the condition of difference always being greater than 1. Different possible scenarios are shown in the figure below



Qn10. Area of the square $= 8 \times 8 = 64cm^2$. There are four triangles which are unshared but part of the square. The area of the fours triangles $= 4 \times 2 = 8cm^2$. Therefore the area of the shaded part $= 64 - 8 = 56cm^2$.

SECTION B

Qn11. (a) At full capacity the tank holds $3 \times 4 \times 5 = 60m^3$ of water. The pipe takes 20min to fill the tank; thus for every minute the pipe puts $\frac{60}{20} = 3m^3$ of water.

(b) A tank measuring $6 \times 12 \times 10$ will hold $6 \times 12 \times 10 = 720m^3$ of water at full capacity. For the pipe which puts $3m^3$ of water per minute it will require $\frac{720}{3} = 240$min $= 4$hours to fill this tank.

Qn12. To catch the thief, distance between the guard and the thief must be zero.
Let x be the distance (in metres) the thief covers before the guard catches him/her; thus the guard will need to run for $(100 + x)$m in order to catch the thief in time t.

Let s_g be the guard's running speed; $s_g = 12km/hr = 200m/min = \frac{10}{3}m/sec.$

Let s_t be the thief's running speed; $s_t = 8km/hr = 400/3m/min = \frac{20}{9}m/sec.$

From $Speed = \frac{Distance}{time}$; $Distance = speed \times time$ and $time = \frac{Distance}{speed}$. Then $x = s_t \times t = \frac{20t}{9}$m.

For the guard; $t = \frac{(100+x)}{s_g}$; substituting for x gives $t = \frac{(100+20t/9)}{10/3} \Longrightarrow t = 30 + 2t/3 \Longrightarrow t/3 = 30 \Longrightarrow t = 30 \times 3 = 90sec = 1.5min$. Thus both the guard and the thief will run for 1min 30sec (1.5min) before the thief is caught.

Qn13. (a) With this clockwise direction numbering, the sum of a pair of flat numbers directly opposite each other is the same for each pair of directly opposite flat numbers. Thus this sum is $46 + 145 = 191$.
The highest numbered flat(max) is opposite flat number 1 and 1+ max $= 191 \Longrightarrow$ max $= 190$. Hence there are 190 flats implying that there are $\frac{190}{2} = 95$ flats on other side of the road. Thus the last flat on the upper row is flat number 95.
Now $95 + X = 191 \Longrightarrow X = 191 - 95 = 96$.

(b) No, the flat number 46 will no longer be directly opposite flat number 145 because with this arrangement, the difference in any pair of flat numbers is 1; with the smaller number being the odd and on the upper side of the road. Therefore the flat number opposite flat number 46 is $46 - 1 = 45$.

Qn14. There are: 9 fiVes, 10 seVens, 10 seVenties; 1 eleVen; 1 twelVe. Thus there are 9+10+10+1+1 = 31 V's and $1 + 1 = 2$ L's. Thus letter V will be written $31 - 2 = 29$ times more than letter L.

Qn15. Starting with order number 0 and following the pattern, order number 0 re-appears after every five operations (order numbers 1 to 5). Thus the 50th operation will have same order as order number 0 and after 52 operations the plate colour on top of the pile is Orange (O).

Order number	0	1	2	3	4	5	...	50	51	52	53	54	55
Arrangement	R	G	O	B	Y	R		R	G	O	B	Y	R
	B	Y	R	G	O	B		B	Y	R	G	O	B
	G	O	B	Y	R	G		G	O	B	Y	R	G
	Y	R	G	O	B	Y		Y	R	G	O	B	Y
	O	B	Y	R	G	O		O	B	Y	R	G	O